Study 2, Qwen2.5-VL-7B: The Full Battery on an A40, the Whole Corpus, and the Unquantised Arm

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Chapter 1

The Full Battery on Scholar's A40, and Stage 6 for the First Time

1.1 Goal

Every Study 2 number so far came from one RTX 3070 with 8 GB of memory, where a single Stage 2 pass takes about ten seconds a question. Stage 6 (stability: five sampled reruns per row) was implemented and tested but never run, because at that speed it would have occupied the card for days. This phase moves the whole battery to Purdue's Scholar cluster (NVIDIA A40, 46 GB), runs every stage there including Stage 6, and checks that the headline figures survive the change of machine.

1.2 What Was Done

Per-machine result trees. A parity run had already shown that greedy decoding reproduces exactly on one GPU but not across GPUs: on identical weights, code, data and pinned libraries the A40 agreed with the frozen 3070 baseline on 96.9% of verdicts and 42.7% of replies byte-for-byte. Every model stage therefore records the GPU in its fingerprint and refuses to resume a file written on another device, and the Scholar run writes to its own tree, `results/scholar-a40/`. Comparisons between stages (Stage 6 against Stage 2, for instance) stay inside one tree, so machine noise is never measured as instability.

Self-continuing job chains. Scholar caps a job at four hours and lets one user run two at once. Each link of a chain queues its own successor first (`afterany`), runs its stages under `timeout --signal=INT` about 25 minutes short of the wall so a stage stops between rows rather than mid-write, and on success marks itself done and cancels the successor. A real failure writes a `FAILED` marker and stops every chain for that model — there is no retry loop. Four chains (Stages 4, 5, 6 and later 8) run after the base chain (Stages 0–3, 7), and the last to finish submits the final chain (Stages 9–11).

1.3 Results

metric	RTX 3070 (frozen)	A40
sensitivity	90.9%	90.4%
specificity	40.9%	40.5%
false-alarm rate	59.1%	59.5%
rule_4 sensitivity ($n = 70$)	30.0%	25.7%
Stage 4 flip rate (black box)	6.4%	6.4%
Stage 5 attention exceeds area null	85.7%	85.4%
Stage 10 explanation correct	2.4%	2.4%

Table 1.1: The same battery on two machines. Every headline agrees within about one point and well inside its 95% interval; rule_4's four-point gap is three images out of 70.

Stage 6, run for the first time (799 rows: 400 baseline-YES, 399 baseline-NO; five reruns each at temperature 0.7):

metric	A40, 4-bit
verdict unanimous across all five reruns	51.9% [48.4, 55.4]
baseline YES / baseline NO	54.2% / 49.6%
majority of reruns overturns the greedy verdict	17.9% [15.3, 20.7]
flickering region (present in some reruns, not all)	38.0% [34.7, 41.4]
region drift > 10% of the image diagonal ($n = 612$)	36.1% [32.4, 40.0]

Table 1.2: Stage 6 stability. Stage 6's own console summary printed a median pairwise agreement of 100%, which reads as stable; unanimity shows half the rows disagree with themselves at least once.

On nearly one row in five, the verdict the paper reports (greedy decoding) is the *minority* answer under the model's own sampling distribution. The explanation is less stable still: where at least two reruns named a region, a third of the time those regions sit more than a tenth of the image diagonal apart.

1.4 Problems and Solutions

Stage 0's small-object score dropped from 89% to 78%. That is 8 of 9 images against 7 of 9 — one image. The verdict becomes "GO with limitation" on the A40, and the paper should not read anything into a one-image difference beyond that.

The "3.5× unexplained latency gap" was the local machine. On the 3070 the same images answered in 3.1 s standalone and 10–11 s inside a run, and the latency caveat in Stage 7 and the runner blamed the harness. On the A40 the in-run median is 2.3 s. The gap belongs to the 3070 machine; the caveat now quotes a standalone-to-in-run ratio only when both were measured on the same GPU.

Region drift was never computed. Stage 6 reads its rows from Stage 2, which carries no image size, so the drift column stayed empty. Stage 3's regions file has the size of every (image, rule), so the report now joins on it and recomputes drift from the stored regions; future Stage 6 runs take the sizes in-run. No other figure moved.

1.5 Standard vs. Non-Standard Choices

Standard. Bootstrap intervals on every rate; resumable, checkpointed runs.

Non-standard. Treating the GPU as part of the experimental configuration — fingerprinted, never mixed within one tree — is stricter than common practice, and it is what the 96.9% parity figure demands: 3.1% of greedy verdicts change with the card alone.

1.6 Implications

The headline results are not an artefact of one machine. Stage 6 adds a new finding for the paper: the greedy verdict is the minority answer in 17.9% of rows, and explanation regions are less stable than the verdicts they accompany.

Chapter 2

Every Compliant Image, Not Only the Sampled Ones

2.1 Goal

The Stage 1 sample is deliberate (architecture document §5.3): all 1,278 images with an annotated violation and 1,278 compliant images matched to them, 3,882 (image, rule) questions in all. That left 7,457 compliant images the model never saw. Specificity measured on the matched stratum is only as good as the matching, so this phase asks the model about every one of them.

2.2 What Was Done

`01_build_sample.py --complement` writes `complement_manifest.csv`: one row per unsampled compliant image, stratum "R", with the rule asked chosen by the same `rules_to_ask` logic as the main sample (one rule per image, as before). The manifest was built locally first (7,457 rows, SHA-256 `cccec48c...`); Scholar rebuilt it and the chain refused to start unless the hash matched, which it did byte for byte. Stage 2 then ran on it in a separate chain, writing `baseline_qwen7b-4bit_complement.csv`, so the frozen sample files are untouched.

2.3 Results

specificity	**estimate [95% CI]**	n
matched compliant stratum (C)	45.1% [42.5, 47.9]	1,278
unsampled compliant images (R)	42.6% [41.5, 43.7]	7,457
all compliant questions	43.0% [41.9, 44.0]	8,735

Table 2.1: Specificity on the whole compliant corpus (A40, 4-bit). The matched stratum is within 2.5 points of the images it was meant to stand for.

rule	**false-alarm rate, all compliant**	n
rule_1 (PPE)	57.5%	2,670
rule_2 (fall protection)	35.4%	2,713
rule_3 (unprotected edge)	93.2%	2,599
rule_4 (machinery zone)	8.5%	753

Table 2.2: False alarms by rule over every compliant question.

2.4 Problems and Solutions

None at run time. The one design choice — a single rule per unsampled image rather than all applicable rules — was kept from the main sample so the two strata are comparable; asking every applicable rule would have roughly doubled the run for a different estimand.

2.5 Standard vs. Non-Standard Choices

Standard. Validating a stratified sample against the population it represents.

Non-standard. Hash-gating the manifest across machines, so the cluster run provably asked the questions that were built and reviewed locally.

2.6 Implications

The study sample does not flatter the model: specificity is about 43% whichever compliant images are used. rule_3 is the extreme — on 93% of compliant questions the model reports a violation — and the paper should state it per rule, since the pooled false-alarm rate hides a range from 8.5% to 93.2%.

Chapter 3

Stage 9 Completeness, Redefined on Attention (F7)

3.1 Goal

Stage 9 (bounded completeness) asks whether an explanation was offered and whether it held up. As designed, "held up" meant the answer flips when Stage 4's token ablation removes the named region. Finding F4 showed that ablation fails its own negative control — a 51.7% artefact floor against a 5.9% signal — so the designed metric correctly computed a quantity with no discriminative power, and was marked not reportable (F7). This phase replaces the broken half.

3.2 What Was Done

The decision (2026-11-10) was to redefine "held up" on attention, which Stage 5 already measures without any intervention and which survived its own null (F6). `09_completeness.py --basis attention` classifies each row:

- **no usable explanation** — no region was named;
- **supported** — the share of attention mass inside the named region exceeds the share of the image the region covers (lift > 1);
- **weak** — a region was named but attention inside it is at or below its area share.

It writes `completeness_attention_{model}.csv`. The ablation basis stays the default so the frozen file reproduces, and its numbers are never reported. The report heads the section "attentional, not causal" and prints the share of regions at a lift of 2× or more beside the headline.

3.3 Results

class	RTX 3070	A40
explanation supported	67.0%	66.5%
explanation weak	11.1%	11.4%
no usable explanation	21.9%	22.2%
lift $\geq 2\times$ (regions only)	43.4%	42.6%

Table 3.1: Stage 9 on the attention basis, $n = 3{,}882$ (regions only: 3,031 / 3,022). Median lift 1.74 / 1.73.

3.4 Standard vs. Non-Standard Choices

Non-standard, and it must be argued in the paper. The claim changes from causal ("removing the region changes the answer") to attentional ("the model looked where it said"). That is a weaker claim, and the paper states it as one. "Supported" is crossed by any lift above 1.0, which is why the 2× line is reported beside it: under half the named regions draw twice their area's share of attention.

3.5 Implications

Completeness is reportable again, on a basis that has passed a null test. A metric defined over another metric inherits that metric's validity; this one now inherits Stage 5's, which was checked, rather than Stage 4's, which failed.

Chapter 4

Stage 8 Robustness at Pre-Approved Strengths

4.1 Goal

Does a verdict, and the region that explains it, survive the kind of image degradation a site camera produces? Stage 8 had not been built. This phase builds it, with the perturbation strengths fixed and approved before any run, so they could not be tuned against results.

4.2 What Was Done

Five perturbations, applied one at a time, never combined:

perturbation	**strength (approved 2026-11-10)**
blur	Gaussian, $\sigma = 1\%$ of the image's long side
low light	-2 EV, applied in linear light, not gamma-encoded values
occlusion	a mean-luminance grey block over the centre 50% $\times$ 50% of the image
contrast	factor 0.4
desaturate	ITU-R 601-2 luma

The subset is 1,600 Stage 2 rows (800 YES, 800 NO, at least 40 per rule), seed 42; each row is asked again under each perturbation with greedy decoding. Per perturbation the stage records the new verdict, whether it matches the baseline, the region, and the baseline-to-perturbed centroid distance over the image diagonal. A surviving verdict whose region moved more than 10% of the diagonal is flagged — the same cut-off Stage 6 now uses. For occlusion it also records how much of the annotated hazard the block hides, so the case where the block covers the hazard itself is reported separately. The strengths are part of the run fingerprint.

4.3 Results

perturbation	verdict survives	survived, region moved $> 10\%$
blur	70.4% [68.2, 72.6]	47.7% ($n = 459$)
low light	90.7% [89.2, 92.1]	13.8% ($n = 884$)
occlusion	80.2% [78.2, 82.1]	35.1% ($n = 643$)
contrast	90.4% [88.9, 91.8]	14.8% ($n = 865$)
desaturate	86.6% [84.9, 88.2]	20.9% ($n = 789$)

Table 4.1: Stage 8, A40, 4-bit, $n = 1{,}600$ per perturbation. The drift column is over rows that survived and named a region both times.

Occlusion that hides at least 10% of the annotated hazard leaves the verdict standing 67.9% of the time ($n = 290$); occlusion that hides less leaves it standing 85.1% ($n = 141$). That is the direction expected if the model's verdict depends on the hazard rather than on the scene around it.

4.4 Standard vs. Non-Standard Choices

Standard. One-factor-at-a-time perturbation; survival rate per perturbation, never pooled.

Non-standard. Strengths fixed and approved before any run; low light simulated in linear light; the occluder's coverage of the annotated hazard recorded per row, so occlusion results separate "hid the evidence" from "hid something else".

4.5 Implications

Mild blur is the most damaging change: three verdicts in ten flip, and when the verdict survives, the region moves in almost half the cases. A surviving verdict is not a surviving explanation — the paper should report both columns.

Chapter 5

The Unquantised Arm: Is 4-bit Compression the Cause?

5.1 Goal

Every Study 2 result used Qwen2.5-VL-7B compressed to 4 bits (nf4) to fit the 8 GB card. A reviewer can ask whether the poor explanation results are a property of the model or of the compression. The A40 has room for the unquantised model in bfloat16, so this phase runs the full battery on it.

5.2 What Was Done

A new tier, `qwen7b-bf16`, loads the same checkpoint folder at the same pinned revision (`cc594898...`) with quantisation off. The chains take the model as a parameter; a non-primary tier gets its own state directory and model-suffixed output files in the same A40 tree, so the two tiers compare without machine noise. Every stage ran: 0–10, plus 8 and the attention basis of 9.

5.3 Results

Table 5.1 sets the two tiers side by side.

Compression changes *what* the model answers — most of all on rule_4 (a worker in the danger zone of heavy machinery), where the unquantised model finds three times as many annotated violations, with intervals that do not overlap. It barely changes *how good the explanations are*: every explanation metric moves by two points or less. The unquantised model is also less steady under sampling, especially on its NO verdicts.

5.4 Problems and Solutions

Stage 6 stopped on its parse-rate guard. After 100 rows only 90% had all five sampled replies readable, below the pre-registered 95% floor. The first hypothesis — the 128-token reply cap cutting the reply off before the final ANSWER line — was wrong: a 200-call diagnostic found the longest reply was 94 tokens. Instead, at temperature 0.7 the model sometimes replies with the single word "NONE" (11 of 13 failures) or "ANSWER: MAYBE". Per reply that is 2.6%; but one bad reply in five failed the whole row.

Decision (2026-11-10): grade the guard per reply, as Stage 2's is. The guard exists to catch a degrading prompt, which a per-reply rate still does. An unreadable reply still counts as a disagreement in the agreement rate, so no 4-bit figure changed. The report gives the unreadable-reply rate on its own line. bf16 Stage 6 was rerun from scratch; the tripped attempt is kept in

metric	**4-bit**	**bf16**
what the model answers		
sensitivity	90.4%	93.8%
specificity	40.5%	36.0%
rule_4 sensitivity ($n = 70$)	25.7% [15.7, 35.7]	81.4% [71.4, 90.0]
Stage 0 small-object ($n = 9$)	78%	89%
how good the explanation is		
Stage 10 explanation correct	2.4%	2.1%
Stage 5 attention exceeds area null	85.4%	85.9%
Stage 9 explanation supported	66.5%	68.3%
Stage 4 flip rate (black box)	6.4%	6.8%
Stage 8 survives blur / occlusion	70.4 / 80.2%	68.8 / 79.4%
stability under sampling (Stage 6)		
verdict unanimous	51.9%	47.0%
baseline NO	49.6%	36.8%
unreadable sampled reply	0.7%	2.3%
in-run median latency	2.34 s	2.02 s

Table 5.1: The same checkpoint, A40, same questions, with and without 4-bit compression.

`results/scholar-a40/superseded/`. This is a change to a pre-registered guard after it tripped, and the paper states it as such.

One Stage 8 row short. bf16 has 1,599 rows, not 1,600: the shared seeded sampler rounds each rule's proportional share, and for bf16's NO verdicts the shares round to 799. The sampler is shared with Stages 4 and 6 and their files are frozen, so it is left as is and noted. One further blur reply out of 7,995 was unreadable.

5.5 Standard vs. Non-Standard Choices

Standard. An unquantised ablation of a quantised model.

Non-standard. Running it through the identical battery on the same GPU, so the comparison isolates compression from machine, data and code.

5.6 Implications

The poor explanation quality reported for Study 2 is not a compression artefact: the unquantised model explains no better. The paper can say so with a measured comparison. Accuracy is a different matter — rule_4's sensitivity under 4-bit is a compression effect and should be reported as one.

Chapter 6

The §5.5 Annotation Audit: Draw and Labelling Page

6.1 Goal

The dataset grades its own annotations, and 47% of images are self-flagged "poor info". Every Study 2 figure is measured against those annotations, so they are a reference standard, not ground truth. Architecture document §5.5 asks a person to relabel 100 images, 50 per quality tier, so that the annotation error rate can be reported beside every result.

6.2 What Was Done

The draw. `12_audit_sample.py` draws from the whole corpus (10,013 images), not the study sample. The pre-registered draw was a pure random 50 per tier with seed 42. It held 3 hazard images in 100, too few to estimate how often a *marked* violation is wrong. **Decided 2026-11-10:** stratify by hazard within tier — 25 images with an annotated violation and 25 without, per tier, each pool drawn with its own seeded generator and independent of record order. Corpus-level error rates are reweighted to the true 12.8% hazard prevalence. The 50 hazard images carry 38 rule_1, 7 rule_3, 3 rule_2 and 3 rule_4 violations, so per-rule rates beyond PPE will rest on very few images. 61 of the 100 images were in the study sample and carry a model answer.

The labelling page. A private web page presents each image, downscaled to a 1,024-pixel long side. Step one is blind: the auditor answers the four rule questions (yes / no / can't tell) from the photograph alone and saves. Only then does the page reveal the dataset's annotation boxes and the model's answer and region, and the blind answers lock. Step two asks, per annotated box, whether it sits on its hazard (yes / partly / no), with a free-text note. Each auditor's answers are stored under their own identity; the owner can read all of them.

Decided 2026-11-11: a frontier model is the first reader. No human auditor was available, and the owner chose to have Claude (Opus 5.5) answer the audit, with the owner reviewing every answer afterwards. The procedure kept the page's blind-then-reveal order. The 100 images were put in a seeded random order (seed 2026) so that the hazard and compliant strata were not grouped. Each image was viewed at the same 1,024-pixel long side and answered for its applicable rules against the dataset's own rule text, with a one-sentence reason, before any annotation was opened. The blind answers were committed to the repository (`7b4adeb`) before the reveal. Only then were the annotation boxes drawn on the 50 hazard images and each judged for placement. Answers are stored in the format the labelling page writes (`results/audit_answers/audit/claude-opus-5-5/`), so the scoring script runs on them unchanged. A second page, `audit/review.tpl.html`, shows each answer beside the annotation, disagreements first, and records the owner's final answer per rule.

measure	first reader	adjudicated
marked, not confirmed	10.0% [2.0, 18.0] (5/50)	2.0% [0.0, 6.1] (1/49)
unmarked, auditor sees one	8.3% [5.3, 11.7] (22/265)	6.4% [3.8, 9.4] (17/266)
can't tell	6.5% [3.9, 9.2] (22/337)	6.5% [3.9, 9.2] (22/337)
image disagrees, reweighted to the corpus	16.9% [8.9, 25.9]	11.9% [5.2, 19.9]
annotated box on its hazard	47 yes, 4 partly, 0 no (n=51)	

Table 6.1: Section 5.5 audit, question level unless stated. First reader: Claude Opus 5.5, blind. Adjudicated: the owner's final answer on the 40 reviewed images, the first reader's on the other 60.

6.3 Results

Table 6.1 gives the scores. The adjudicated column is the one the paper reports; the first-reader column is kept so the effect of the review is visible.

6.3.1 First reader

Marked, not confirmed (5 of 50). Each was looked at after the reveal. One is the reader's own miss: two workers in shorts (image 0001305), a rule_1 violation under the dataset's rule text that the reader overlooked. One is a filing difference: the reader called two workers climbing a steep slope a fall-protection case, and the annotation called it an unguarded edge (0000991). One is a distant figure the reader saw as helmeted (0002260). Two are rule_3 judgements where the reader saw a graded slope or a fence along the rim (0004032, 0007712).

Unmarked, auditor sees one (22 of 265). Half are rule_4. Eleven times the reader saw a worker within a few metres of an operating excavator that the annotation left unmarked. Nine of those were on images annotated only for another rule, usually a missing hard hat on the same worker. Next come rule_3 (6), rule_2 (3) and rule_1 (2). The annotation looks reliable for PPE and incomplete for plant proximity.

Box placement. No annotated box was off its hazard. Four were partly off, all rule_1 boxes drawn around a group that included helmeted workers.

6.3.2 Owner review (2026-11-12)

The owner opened the 40 images the review page put first: the 24 where the first reader disagreed with the annotation and the 16 where it agreed but answered "can't tell" somewhere. The other 60, where the first reader agreed with the annotation on every rule, were accepted without being opened. Reviews were exported from the page (`results/audit_review/`) and merged by `14_audit_adjudicate.py`, which writes the adjudicated answers (`results/audit_answers_adjudicated/`) for the unchanged scorer.

Agreement. On the 139 rule questions of the reviewed images the owner kept the first reader's answer 127 times (91.4%) and changed 12, on 10 images. These 40 are the hardest images by construction, so the figure is a lower bound for agreement over all 100. Every one of the 12 changes moved towards the annotation.

- *Marked, not confirmed, overturned (3).* Image 0001305 (shorts), the reader's own miss. Image

0002260: the distant worker wears a cap, not a hard hat (owner about 80% sure). Image 0007712: the green guardrail stops short along the rim, so part of the edge is unguarded.

- *Unmarked violations withdrawn (5).* Two rule_4: in 0010538 the worker only looks inside the excavator's radius because of the camera angle, and in 0010327 it is not clear the machine is operating. Three slope calls: in 0000991 (rule_2) and 0011925 (rule_3) the ground is a steep graded slope, not a drop, and in 0004657 (rule_3) the left side is guarded and the right is graded.
- *Can't tell resolved (4).* Three "can't tell" became No (0000991 rule_1, 0009735 rule_3, 0011292 rule_1); one No became "can't tell" (0000991 rule_3), and in total "can't tell" did not change.

Do graded slopes count? Decided 2026-11-12: no, unless there is a drop. The owner's note on 0000991 asked whether a graded slope falls under the fall-protection rule at all and suggested checking OSHA. The governing text is the dataset's own. Rule 2 needs work "from a height of three meters" where "the edges are without any edge protection". Rule 3 needs "underground projects three meters in depth with steep retaining wall". Both describe an edge or a near-vertical wall, not a slope you can walk down. OSHA construction fall protection (29 CFR 1926 Subpart M) is built the same way. Its trigger is an "unprotected side or edge" of a walking/working surface 6 ft (1.8 m) or more above a lower level (1926.501(b)(1)), and the only sloped surfaces it names are roofs (1926.501(b)(11), steeper than 4 in 12). No OSHA interpretation letter on earth slopes or embankments was found. The OSHA rules for excavation slopes (Subpart P, Appendix B: at most 34–53° by soil type for excavations under 20 ft) concern cave-in, not falls. Excavation edges need guarding only when the excavation is hard to see, or is a well, pit or shaft (1926.501(b)(7)). The audit therefore reads a walkable graded slope with no drop-off as no rule 2 or rule 3 violation. A sheer drop at the top or bottom of the slope still counts. That supports the owner's calls on 0000991, 0011925 and 0004657, and the one remaining "marked, not confirmed" (0004032, a rule 3 box on a graded slope) as an annotation error. The images come from a Chinese site corpus and the rules are the dataset's, not OSHA's. OSHA is cited only to show that a regulator draws the same line, and the paper says so.

After adjudication. One marked violation is not confirmed: 0004032, a rule_3 box where reader and owner both see a graded slope. Seventeen unmarked violations stand: rule_4 9, rule_3 4, rule_2 2, rule_1 2. Eight of the nine rule_4 cases are on images annotated for another rule, and 11 of the 50 hazard images carry at least one violation the annotation does not mark.

What the review says about the first reader. Its errors run one way. It over-calls the two rules that need a judgement of three-dimensional space from a flat photograph: proximity to an excavator, where camera angle shortens distances, and slopes, which it read as unguarded drops. On PPE it was wrong twice: a distant cap it read as a hard hat, and the shorts clause of rule_1, which it under-applied.

6.4 Scoring

The scoring script, `13_audit_score.py`, was written and tested ahead of the answers. It keeps two errors apart — *marked, not confirmed* (the annotation marks a violation the auditor does not see) and *unmarked, auditor sees one* — reports "can't tell" on its own line, and scores only rules that apply to the image. Its corpus figure reweights the 50/50 draw to each tier's real hazard prevalence (rich info 771 of 5,271 images, poor info 507 of 4,742) and the tiers to their real shares, with a bootstrap that resamples images within each (tier, stratum) cell. Per tier, "marked, not confirmed" rests on about 25 images and falls below the $n = 30$ reporting floor by design, so it is quoted pooled across tiers.

	4-bit		bf16	
	annotation	adjudicated	annotation	adjudicated
sensitivity	87.8% [77.6, 95.9]	82.1% [71.4, 91.1]	91.8% [83.7, 98.0]	92.9% [85.7, 98.2]
specificity	43.9% [31.6, 56.1]	42.0% [28.0, 56.0]	33.3% [21.1, 45.6]	38.0% [24.0, 52.0]
accuracy	64.2% [55.7, 72.6]	63.2% [53.8, 72.6]	60.4% [50.9, 69.8]	67.0% [58.5, 75.5]
violations	49	56	49	56

Table 6.2: Both tiers on the 106 audit questions, scored against each reference. These are rates as drawn (25 hazard and 25 compliant images per quality tier), not corpus estimates.

6.5 The Models Against the Adjudicated Labels

Sixty-one of the 100 audit images are in the study sample, carrying 112 model questions between them. `19_audit_rescore.py` scores both tiers on those questions twice: once against the annotation, once against the adjudicated labels. Six questions the adjudicated reader could not tell are dropped from both, so the two scores use the same 106 questions (`results/audit_rescore.txt`).

The two references disagree on 9 questions: 8 violations the annotation leaves unmarked (five rule_4, three rule_3) and one marked rule_3 violation the adjudication rejects. The unquantised model sides with the adjudicated label on 8 of the 9. The 4-bit model sides with it on 4, and misses four of the five unmarked rule_4 violations. So correcting the reference moves the two tiers apart. Against the annotation the 4-bit model is slightly more accurate (64.2% against 60.4%). Against the adjudicated labels the unquantised model is (67.0% against 63.2%). Each interval spans about 18 points, so the direction of the change is the finding, not its size. It agrees with the rule_4 audit (Chapter 7): the annotation's omissions penalise the more sensitive tier.

6.6 Standard vs. Non-Standard Choices

Standard. Blind relabelling of a reference standard; stratified sampling with reweighting.

Non-standard, and stated in the paper. The stratified draw replaces the pre-registered pure random draw, decided after seeing that the random draw could not answer the question it was for. Blind-then-reveal in one page keeps the first answers uncontaminated while still collecting box-level judgements.

Non-standard, and stated in the paper. The first reader is a vision-language model, not a person. It is a different and larger model family than the one under study, but it can share that kind of model's blind spots, chiefly distant, few-pixel workers (10 of its 22 "can't tell" answers were on such figures). The owner's review is what makes the audit a human one, and it is partial by design: the owner opened the 40 images where the reader disagreed with the annotation or could not tell, and accepted the 60 where reader and annotation agreed. An error the reader and the annotation share would survive that review. The paper reports the first reader's figures, the adjudicated figures, and the 91.4% agreement between reader and owner on the images the owner opened.

6.7 Implications

Every accuracy and explanation figure in Study 2 is "agreement with the dataset's annotations". After adjudication about one corpus image in eight (11.9% [5.2, 19.9]) disagrees with its annotation. Almost all of that is violations left unmarked; a marked violation is rarely wrong (1 of 49).

- **The annotations can be trusted where they mark something.** Marked violations are confirmed 48 of 49 times and every box is on or partly on its hazard. Low explanation IoU (F3) is therefore not a symptom of badly placed annotation boxes. It was a coordinate-frame error in reading the model's boxes (Chapter 8), and corrected, 55% of correct Yes answers overlap the annotated box at IoU 0.3 or more.
- **They are incomplete, mostly for rule_4.** Nine of the 17 unmarked violations are a worker near an operating excavator, and 11 of 50 hazard images carry a second, unmarked violation. So F3's third caveat (a zero-IoU region may be a real, unannotated hazard) is real but bounded. Some of the model's rule_4 "false alarms" may be real hazards, and the unquantised model's higher rule_4 sensitivity is measured against an incomplete rule_4 reference. No frozen number changes; both readings are stated beside the rule_4 results.
- **The incompleteness estimate is the soft one.** It rests on the first reader seeing a violation and the owner keeping it, and both reviews showed that rule_4 and slope judgements from a photograph are the least certain calls. It can err both ways: some of the 17 may still be perspective effects, and the owner did not search the 60 unopened images for violations both reader and annotation missed.

Chapter 7

The Rule 4 Audit: Are the Excavator False Alarms Real?

7.1 Goal

The §5.5 audit (Chapter 6) found that rule 4 is the rule the annotation most often leaves unmarked: a worker within reach of an operating excavator. Every rule 4 accuracy figure in Study 2 is scored against that incomplete reference. Two readings depend on it. The unquantised model's rule 4 "false alarms" (31% of compliant images) may be real hazards. Its much higher rule 4 sensitivity (81% against 26% for 4-bit) is measured against the same incomplete reference. This chapter relabels a stratified draw of the sample's rule 4 questions and corrects both tiers' rule 4 confusion matrix.

7.2 What Was Done

The draw (decided 2026-11-12, committed as `1973153` before any image was viewed). The frozen sample holds 323 rule 4 questions, one per image. `15_rule4_audit_sample.py` splits them into nine cells by stratum and by what each tier answered:

- C: compliant images;
- H: hazard images where rule 4 is not annotated;
- P: images where rule 4 is annotated.

Each cell is drawn with its own seeded generator (12 per negative cell, 8 per positive cell; small cells whole), giving 92 images. The positive cells are in the draw so the reader cannot infer the answer from the draw itself, and they estimate how often a marked rule 4 violation is wrong.

The reader. As in Chapter 6, Claude (Opus 5.5) answered first. It answered rule 4 only, in a seeded random order, against the dataset's rule text, without seeing the cell, the annotation, or either model's answer. The answers were committed (`687059b`) before scoring. The standard applied the owner's two corrections from the §5.5 review:

- camera angle shortens distances;
- the excavator must be working or visibly manned.

Where either could not be judged, the answer is "can't tell".

The correction. `16_rule4_audit_score.py` takes, per cell, the share of answered rows the reader calls a violation and gives every one of the 323 rows its cell's share as the probability that it is a real violation. Each tier's confusion matrix is then recomputed in expectation, beside the annotation-based one, with a bootstrap over answered rows within each cell.

cell	size	drawn	says violation	first reader
C, both tiers No	85	12	1 / 12	1 / 12
C, 4-bit No, bf16 Yes	32	12	5 / 10	7 / 10
C, 4-bit Yes	8	8	3 / 7	3 / 7
H, both No	28	12	2 / 9	2 / 9
H, 4-bit No, bf16 Yes	59	12	6 / 12	5 / 10
H, both Yes	41	12	4 / 9	7 / 11
P (annotated), all cells	70	24	18 / 18	18 / 18

Table 7.1: Rule 4 audit by draw cell, adjudicated, beside the first reader alone. "Can't tell" (15 of 92 in both) is excluded from the share.

	4-bit		bf16	
	annotation	corrected	annotation	corrected
sensitivity	25.7%	26.4% [18.3, 34.5]	81.4%	82.2% [72.8, 90.0]
specificity	80.6%	84.1% [77.1, 91.4]	45.1%	58.1% [49.8, 68.2]
false alarm, compliant images	6.4%	4.6% [1.3, 7.6]	31.2%	20.3% [10.8, 28.6]
false alarm, other-hazard images	32.0%	30.8% [15.1, 45.7]	78.1%	70.6% [56.9, 81.1]

Table 7.2: Rule 4 over the 323 sample rows, scored against the annotation and corrected by the adjudicated relabels (`results/rule4_audit_score_owner-adjudicated.txt`). The first reader alone had put bf16's compliant-image false-alarm rate at 14.8% and its specificity at 63.3%.

7.3 The Owner's Review

The owner reviewed every image where the first reader's answer disagreed with the annotation or was "can't tell": 40 of the 92. The other 52 agree with the annotation and keep the first reader's answer. `18_rule4_adjudicate.py` merges the review, with the owner's answer taking precedence. The owner changed 8:

- *First reader Yes, owner No or can't tell (5 of 25).* The worker is outside the radius (0007497, 0010299) or probably outside it (0004350). The excavator may not be working (0010103). One note asks whether a mini excavator is a threat at all (0012295).
- *First reader can't tell, owner decided (3 of 15).* Two No (0001436, 0003886), one Yes (0002534: within four or five metres of the hammer).

The owner kept 20 of the first reader's 25 unannotated Yes calls. Twelve images stay "can't tell" for both readers. On one (0005496) the operator's cab is out of frame, so whether it is manned cannot be judged. On another (0010267) the workers stand above the lowered bucket, a case the rule's text does not cover.

7.4 Results (adjudicated)

The annotation misses more rule 4 violations than it marks. Every annotated rule 4 violation that could be judged was confirmed (18 of 18). The adjudicated labels put about 80 real violations among the 253 unannotated rows, against 70 annotated. The first reader alone had estimated 95. Most of them are on images where the unquantised model said Yes: 5 of 10 such compliant images and 6 of 12 such other-hazard images.

What changes for the two tiers.

- *bf16.* About a third of its "false alarms" on compliant images are real hazards. Its corrected false-alarm rate there is 20.3%, not 31.2%, and its rule 4 specificity rises from 45% to 58%.
- *4-bit.* Its rule 4 blindness is real. Corrected sensitivity is 26%, and the gap to bf16 (82%) is unchanged.

So quantisation costs rule 4 sensitivity, as Chapter 5 found, and a substantial part of what looked like the unquantised model's over-triggering is really the annotation's omission. The owner's review moved the figures back toward the annotation without changing either conclusion.

Test–retest. Four images were also in the §5.5 draw. The reader gave the same rule 4 answer on three. On the fourth (0009901) it moved from Yes to "can't tell", applying the stricter standard adopted after the owner's review.

7.5 Standard vs. Non-Standard Choices

Standard. Stratified relabelling with design weights. Strata fixed and committed before any image was viewed.

Non-standard, and stated in the paper.

- The cells are defined by the models' answers, which is what lets the correction apply to each tier. The reader was kept blind to the cells for exactly this reason.
- The correction assumes that the share of real violations is the same for every row in a cell. Cells of 8 to 12 answered rows make that share coarse, so the intervals are wide.
- The first reader is a vision-language model. The owner reviewed only the 40 images where it disagreed with the annotation or could not tell; the 52 where it agreed were not re-read. The owner kept 20 of its 25 unannotated Yes calls, close to the §5.5 review, where 9 of 11 were kept.
- Fifteen images (16%) stay "can't tell". They are left out of each cell's share, which amounts to assuming they are violations at the same rate as the answered rows.

7.6 Implications

The rule 4 row of every Study 2 table reads differently once the reference is corrected. The 4-bit model's near-blindness to rule 4 (F2) stands. The unquantised model's rule 4 false-alarm rate on compliant images is about two thirds of what the annotation says (20% against 31%). More generally, a model that is more sensitive than the annotators is penalised as a false-alarm generator, which matters for how the paper reads F1 ("accuracy is bought with false alarms").

Chapter 8

The Region Frame: Reading the Model's Boxes Where It Drew Them

8.1 Goal

Every region-based figure in Study 2 rests on reading the model's box correctly. While preparing the model roster, a check of Qwen2.5-VL's documentation raised a question: does the model state its box in the original photograph's pixels, or in the pixels of the resized image it was actually given? This chapter records the answer, the fix, and every figure the fix changes.

8.2 What Was Found

Qwen2.5-VL resizes each image to fit the pixel budget (640 blocks of 28×28, about 0.5 megapixels) before it sees it. A 1200×900 photograph becomes 812×588. The model states its box in that resized frame. The harness read the box as original-image pixels, so every box was shrunk toward the top-left corner by the resize factor, which was about 0.7 for most of the sample. Three independent checks agreed:

- *Edges.* On 1200×900 images the largest right edge the model ever wrote is 839, just past the resized width of 812 and far short of 1200.
- *Overlap.* Mapping the boxes back to the original frame raises mean IoU with the annotated boxes from 0.058 to 0.389 (4-bit) and from 0.060 to 0.398 (bf16). IoU ≥ 0.5 goes from 2.1% to 40.6% (`results/diag_region_frame.txt`).
- *Processor.* Running the model's own image processor on a 1200×900 blank gives a 29×21 patch grid, i.e. 812×588 pixels: exactly the frame the boxes fit.

Why the earlier check missed it. Before F3 was reported, the highest-IoU rows were checked by hand against the annotation and matched to three decimals. That check was sound for the rows it looked at, and those rows were the problem. Eighteen of the 25 rows with IoU ≥ 0.5 were images already within the pixel budget, where the resize factor is about 1 and the two frames coincide. Such images are about 5% of the sample. Checking only the rows that looked right selected exactly the rows the error did not touch. A coordinate check needs rows chosen without regard to their score.

8.3 What Was Done

The fix, behind a flag. `xai_vlm/frames.py` maps a box from the model-input frame to the original image, using the patch grid the processor reports for that image (`image_grid_thw`, 14-pixel patches). The backend applies it when `XAI_REGION_FRAME=rescaled`. The default, `legacy`, reads boxes as before, so every frozen file still reproduces byte-for-byte. A rescaled run records the flag in its stage

fingerprints and must write to its own results tree. An empty value counts as unset; the first validation job failed on exactly that and is kept in the log.

Stages that only read regions: no GPU. `17_region_frame_fix.py` builds `results/scholar-a40-rf/` from the frozen A40 tree without touching it:

- Stage 2 regions are re-parsed from each stored reply in the input frame, then mapped. Nothing is lost or gained: 4-bit 3,022 mapped and 860 empty, bf16 3,086 and 796.
- Stage 6 rerun regions and Stage 8 perturbed regions are mapped; drift is recomputed.
- Stages 3, 7 and 10 then re-run against the new tree.

Stages that used the regions on the GPU. Stage 4 masks the region and Stage 5 measures attention inside it, so both re-ran on the A40 for both tiers under the flag, followed by Stages 9, 10 and 11.

Validation. Before the reruns, `cluster/rf_validate.sbatch` re-asked 40 rows with the backend mapping boxes itself. All 40 replies matched the frozen replies word for word, and all 40 mapped boxes matched the offline mapping (`RF_VALIDATE PASS`, job 488128). Every new Stage 4 row uses exactly the corrected Stage 2 region, and Gini, which does not depend on the region, is unchanged (median 0.555 in both trees).

Two small gaps. Stage 10 re-ran on the GPU node: two 4-bit rows and one bf16 row differ from the CPU build in the fourth decimal of text similarity, a numerical difference with no effect on any verdict. Stage 5 was meant to save its attention grids so that later region changes could be re-scored without a GPU. The cluster copy predated that change, so no grids were saved. The figures are unaffected.

8.4 Results

The model locates hazards far better than reported. When it correctly says Yes, its box overlaps the annotated box by IoU 0.3 or more 55% of the time, and a third of its explanations are fully correct (right hazard, similar reason, overlapping box). F3's headline of 2.4% was the coordinate error, not the model. The weak point is now split between the box and the wording: 56% of reasons pass the similarity bar and 55% of boxes pass the IoU bar. Poor-info images score higher than rich-info ones (43% against 24% for 4-bit), a split that needs explaining.

Attention sits inside the box. Median attention mass inside the stated region is ten times the region's share of the image (23.8% against 2.4%), and it beats the area null on 97% of rows. On the attention basis, 78% of explanations have a lift of 2× or more, against 43% before. The old tension between F3 and F6 (attention on target, boxes off target) disappears: the boxes were on target and the harness misplaced them.

Masking the region now moves the answer, but only some rules clear the artefact floor. Blacking out the corrected region flips a quarter of Yes answers (25.8% [24.2, 27.5] for 4-bit), four times the old figure. The black-mask floor is 24.4% [20.4, 28.7], so overall the signal and the floor still overlap. By rule they separate:

- rule_4: 65.2% [53.0, 75.8] (4-bit) and 54.6% (bf16), well above the floor;
- rule_2: 45.5% [40.2, 50.8] and 40.8%, above it;
- rule_1: 30.6% [28.0, 33.2] and 28.7%, at the floor;
- rule_3: 9.8% and 15.6%, below it.

	4-bit		bf16	
metric	**before**	**after**	**before**	**after**
Stage 10: is the explanation correct? (true YES rows)				
explanation correct	2.4%	31.6%	2.1%	33.8%
IoU ≥ 0.30	6.7%	54.7%	6.6%	55.8%
median IoU	0.00	0.37	0.00	0.39
right answer, wrong reason	97.6%	68.4%	97.9%	66.2%
Stage 4: does masking the region flip the answer?				
flip, black mask (YES rows)	6.4%	25.8%	6.8%	27.9%
artefact floor, black (NO→YES)	27.2%	24.4%	31.6%	33.8%
flip, token ablation (YES rows)	6.1%	15.8%	5.2%	16.1%
artefact floor, tokens (NO→YES)	51.1%	40.3%	58.2%	45.1%
Stage 5: is attention inside the region?				
attention exceeds area null	85.4%	97.2%	85.9%	97.6%
median attention mass inside	2.4%	23.8%	2.2%	23.6%
median region area	1.1%	2.4%	1.0%	2.1%
Stage 9: completeness, attention basis				
explanation supported	66.5%	75.7%	68.3%	77.6%
lift $\geq 2\times$ (regions only)	42.6%	77.5%	43.2%	78.7%
Stages 6 and 8: does the region stay put?				
rerun drift $> 10\%$ of diagonal	36.1%	56.2%	38.2%	55.4%
blur: verdict holds, region moves	47.7%	61.2%	42.6%	57.4%
occlusion: verdict holds, region moves	35.1%	46.3%	34.2%	46.2%

Table 8.1: Every region-based figure, A40 tree before (`scholar-a40`) and after (`scholar-a40-rf`) the fix. Verdict-only figures (sensitivity, specificity, stability of the verdict, survival under perturbation) do not change. Intervals are in `results/scholar-a40-rf/report_<tier>.txt`.

Token ablation still has a floor (40.3%) far above its signal (15.8%), so the token path remains uninformative.

Regions are less stable than reported. Measured in the right frame, boxes move further. Between greedy and sampled reruns, 56% drift by more than a tenth of the image diagonal (36% before). Under mild blur the verdict often holds, but the box moves that far 61% of the time.

8.5 Standard vs. Non-Standard Choices

Standard. Reading a model's coordinates in the frame it was given; Qwen2.5-VL's documentation uses absolute coordinates in the resized input.

Non-standard, and stated in the paper.

- The fix is behind a flag and the frozen trees are kept, so every pre-fix number can still be reproduced and compared. The paper reports the corrected tree.
- The stages that only read regions were corrected offline from stored replies rather than re-run. That is exact for Stage 2 (the reply is kept) and was confirmed against a fresh GPU run on 40 rows.

8.6 Implications

- **F3 reverses.** The model does not "describe hazards well and locate them badly". It locates the annotated hazard about half the time when it is right, and its full explanation is correct about a third of the time. "Right answer, wrong reason" falls from 98% to 68%: still the majority, but no longer near-total.
- **F4 softens and becomes rule-specific.** The probe is not drowned everywhere. For struck-by (rule_4) and fall protection (rule_2), masking the stated region removes the answer well beyond what masking does to answers that had nothing to find. For PPE it does not separate, and for edge protection it is below the floor.
- **F6 strengthens, and F7's lift caveat is gone.** Attention is inside the box on 97% of rows at a median lift of 8× (4-bit) and 9× (bf16); over three quarters of regions are at 2× or better.
- **Region stability is a new weakness.** The box is right more often than reported, and less steady than reported.
- **For the model roster.** Every new model states boxes in its own convention (resized pixels, 0–1000, 0–1, points). Each adapter will map to the original frame and be checked on rows drawn without regard to their score, before any metric is computed.

Chapter 9

One Backend for Every Model

9.1 Goal

The model roster (fifteen models, approved 2026-11-12) is only a comparison if every model is measured by the same instrument. Until now the harness had one backend, written for Qwen2.5-VL. This step splits it in two: a shared backend that holds every piece of measuring logic once, and a small adapter per model family that holds only what genuinely differs. The test of the step is that the shared backend, given the Qwen2.5-VL adapter, reproduces the frozen backend exactly.

9.2 What Was Done

The split. `backends/hf.py` (`HFVLMBackend`) holds generation and timing, reply parsing, black masking, token ablation and the attention read-out. `backends/families.py` holds one adapter per family, with five duties:

- loading the model and its processor;
- turning one image and one prompt into model inputs;
- stating the frame a pixel box is given in (for models that answer in pixels);
- locating the image tokens and the grid they form;
- merging vision features into the token embeddings, for token ablation.

A family whose image tokens do not form one grid over the whole image (tiled encoders, or Llama 3.2's cross-attention) reports no grid. Attention and token ablation are then marked not applicable for that model rather than computed on a guessed layout, as the roster design requires (D4).

The logic was moved, not rewritten. The token-cell test and the attention average were lifted out of the frozen backend into two functions, `cells_to_keep` and `aggregate_attention`. CPU tests check both against verbatim copies of the frozen loops: 200 random regions on three grid shapes for the first, synthetic attention tensors for the second (exact equality).

Coordinate conventions. `frames.parse_in_convention` reads a reply's REGION in one of four units and returns it in original-image pixels:

- `input_px`: pixels of the resized input (Qwen2.5-VL); exactly the corrected path of Chapter 8;
- `norm1000`: 0–1000 over the image (Qwen2-VL, Qwen3-VL, InternVL, GLM);
- `norm1`: 0–1 over the image (the LLaVA family);

tier	rows	reply	region	black	tokens	attention	max $\lvert\Delta\rvert$
qwen7b-4bit	40	40	40	40	40	40	0
qwen7b-bf16	40	40	40	40	40	40	0

Table 9.1: The shared backend against the frozen one on the A40 (job 488517, commit `89950a9`, `XAI_REGION_FRAME=rescaled`). Counts are rows identical in each field; max $\lvert\Delta\rvert$ is the largest difference between the two attention grids. Rows cover all four rules (4-bit: 16, 7, 14, 3; bf16: 20, 5, 10, 5), both verdicts, and about a third of rows whose black-masked reply differs from the clean one, so the intervention paths were exercised, not only the clean one. Files: `results/backend-parity/`.

- `none`: verdict-only models; the region is dropped.

Each has a round-trip test on one known box. The prompt follows the convention (roster D2): only the REGION line changes, and a test checks that the default prompt is byte-identical to the frozen one for every rule.

Behind a flag. `XAI_BACKEND` selects the class for the two frozen tiers. The default, `legacy`, keeps the frozen `QwenBackend`, so nothing already reported can move; `hf` routes them through the shared backend. New tiers have no frozen results and always take the shared backend, and they are always read in their own convention whatever `XAI_REGION_FRAME` says. Every model stage now gets its backend from one factory (`backends/load.py`), and its fingerprint records the class and convention whenever it is not the frozen path.

Registry. Each tier now names its family, its box convention and its environment (A, the frozen `transformers` 4.49; B, the current release). Two Qwen2.5-VL size arms are registered and pinned: `qwen3b-bf16` and `qwen32b-4bit`. Llama 3.2-11B-Vision is downloaded (Meta approved access on 2026-11-13) and pinned; it gets its adapter at its own Stage 0.

9.3 Results

The check. `scripts/20_backend_parity.py` runs the same rows through both backends, one after the other, in one process on one GPU. Rows are drawn at random (seed 0) from those with a stated region, without regard to their score, as Chapter 8 requires. For each row it records:

- the reply to the clean image, its verdict and its region;
- the reply with the region blacked out (Stage 4);
- the reply with the region's image tokens removed (Stage 4);
- the attention grid (Stage 5) and the patch grid.

Replies are compared as text, so "identical" means byte for byte.

Every field matched on every row, for both tiers (Table 9.1). The attention grids are equal to the last bit. A smoke run of eight 4-bit rows on the RTX 3070 gave the same result before the cluster run.

One false start. The first cluster job (488516) stopped at its GPU-test gate. The new GPU tests for the shared backend run before the frozen backend's 8 GB budget test, and that test reads the process-wide peak memory. It was measuring the previous test's model. The shared-backend tests now

free their model and reset the peak counter when they finish. No backend code changed between the two jobs.

9.4 Standard vs. Non-Standard Choices

Standard. A model-agnostic harness with per-model adapters is how evaluation suites such as VLMEvalKit and lmms-eval are built.

Non-standard, and stated in the paper.

- The frozen backend is kept beside the shared one instead of being replaced, so the frozen results are reproducible from the code that made them. The parity check is the evidence that the two agree.
- Each model is asked for REGION in its own trained convention, not in one common unit. The question and the rest of the format are identical.

9.5 Implications

- Adding a model is now writing one adapter of five duties. Nothing that computes a metric is touched.
- The coordinate error of Chapter 8 cannot recur silently: every tier states its convention, every convention has a round-trip test, and every new adapter is checked on rows drawn without regard to their score before its first metric.
- Next, queued on Scholar behind this check: the full battery for `qwen3b-bf16`; Stage 0 sizing for `qwen32b-bf16` (two A40s) and `qwen32b-4bit`; the verdict-only battery for Llama 3.2 Vision; and the parity run of `qwen7b-bf16` in environment B (transformers 5.20.2, roster D5).